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#include <stdio.h>

int main() {

    int n, tq;
    int at[20], bt[20], rt[20];
    int wt[20], tat[20], rt_time[20];
    int time = 0, completed = 0;

    float avg_wt = 0, avg_tat = 0, avg_rt = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(int i=0; i<n; i++){
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);

        rt[i] = bt[i];
        rt_time[i] = -1;
    }

    printf("Enter Time Quantum: ");
    scanf("%d", &tq);

    while(completed < n){

        int executed = 0;

        for(int i=0; i<n; i++){

            if(at[i] <= time && rt[i] > 0){

                if(rt_time[i] == -1)
                    rt_time[i] = time - at[i];

                if(rt[i] > tq){
                    time += tq;
                    rt[i] -= tq;
                }
                else{
                    time += rt[i];
                    rt[i] = 0;

                    tat[i] = time - at[i];
                    wt[i] = tat[i] - bt[i];
                }
            }
        }
    }
}

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        avg_wt += wt[i];
        avg_tat += tat[i];
        avg_rt += rt_time[i];

        completed++;
    }

    executed = 1;
}

if(executed == 0)
    time++;
}

printf("\nPID\tAT\tBT\tWT\tTAT\tRT\n");

for(int i=0;i<n;i++){
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,at[i],bt[i],wt[i],tat[i],rt_time[i]);
}

printf("\nAverage Waiting Time = %.2f",avg_wt/n);
printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
printf("\nAverage Response Time = %.2f\n",avg_rt/n);

return 0;
}

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*C:\Users\Admin\Desktop\ x + v
Enter number of processes: 3
Enter Arrival Time and Burst Time for P1: 1 2
Enter Arrival Time and Burst Time for P2: 3 4
Enter Arrival Time and Burst Time for P3: 4 5
Enter Time Quantum: 3

PID  AT   BT   WT   TAT  RT
P1    1    2    0    2    0
P2    3    4    3    7    0
P3    4    5    3    8    2

Average Waiting Time = 2.00
Average Turnaround Time = 5.67
Average Response Time = 0.67

Process returned 0 (0x0)  execution time : 14.072 s
Press any key to continue.
|
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